

Willowbrook General Hospital

Network Build Record, IP Addressing, Access Controls & Lessons Learned

1. Purpose of This Document

This record captures the network as it was actually built and configured in Packet Tracer, including the exact IP addressing assigned to every interface and site, the access control (ACL) permissions applied and the reasoning behind each one, and the key lessons learned while troubleshooting the build. It is a companion to the original design document and reflects the final, verified state of the network rather than the initial plan, a small number of details (ACL naming, one abandoned control) changed during implementation, and those changes are documented here along with the reasoning.

Scope: one main hospital campus and two satellite clinics (Clinic A and Clinic B), connected via point-to-point WAN links, running multi-area OSPF, with departmental VLANs, guest Wi-Fi, a DMZ-hosted patient portal, and layered ACL security.

2. IP Addressing - WAN Links

Link	Subnet	R-MAIN Side	Remote Side
R-MAIN ↔ R-ClinicA	172.20.255.0/30	172.20.255.1 (Se0/1/0, DCE)	172.20.255.2 (Se0/1/0, DTE)
R-MAIN ↔ R-ClinicB	172.20.255.4/30	172.20.255.5 (Se0/1/1, DCE)	172.20.255.6 (Se0/1/0, DTE)
R-MAIN ↔ R-ISP	198.51.100.0/30	198.51.100.1 (Se0/2/0, DTE)	198.51.100.2 (Se0/1/0, DCE)

3. IP Addressing - Main Hospital (R-MAIN, Area 0)

VLAN	Department	Subinterface	Subnet	Gateway
10	Reception / Admissions	Gi0/0/0.10	172.20.10.0/24	172.20.10.1
20	Doctors	Gi0/0/0.20	172.20.20.0/24	172.20.20.1
30	Nursing	Gi0/0/0.30	172.20.30.0/24	172.20.30.1
40	Pharmacy	Gi0/0/0.40	172.20.40.0/24	172.20.40.1
50	Radiology / Lab	Gi0/0/0.50	172.20.50.0/24	172.20.50.1
60	Administration (IT)	Gi0/0/0.60	172.20.60.0/24	172.20.60.1
70	Guest Wi-Fi	Gi0/0/0.70	172.20.70.0/24	172.20.70.1
90	DMZ — Patient Portal	Gi0/0/0.90	172.20.90.0/24	172.20.90.1

200	Servers — EHR (internal)	Gi0/0/0.200	172.20.200.0/24	172.20.200.1
99	Native / switch mgmt	(trunk native)	-	-

Key hosts:

- Server-EHR: 172.20.200.10 /24, gateway 172.20.200.1 (static, referenced directly by IP in the EHR-ACCESS ACL)
- Server-Portal: 172.20.90.10 /24, gateway 172.20.90.1 (static, referenced directly by IP in the DMZ ACLs)
- Department PCs and Laptop-Guest: DHCP, pulled from the pool matching their VLAN (e.g. Laptop-Guest received 172.20.70.2)
- OSPF Router ID: 1.1.1.1 - backbone router, all local VLANs and both clinic-facing serial links sit in Area 0

4. IP Addressing - Clinic A (R-ClinicA, Area 1)

VLAN	Department	Subinterface	Subnet	Gateway
10	Reception	Gi0/0/0.10	172.20.100.0/24	172.20.100.1
20	Doctors	Gi0/0/0.20	172.20.101.0/24	172.20.101.1
30	Nursing	Gi0/0/0.30	172.20.102.0/24	172.20.102.1
70	Guest Wi-Fi	Gi0/0/0.70	172.20.103.0/24	172.20.103.1
99	Native / switch mgmt	(trunk native)	-	-

- Serial0/1/0: 172.20.255.2/30 (DTE) toward R-MAIN
- OSPF Router ID: 2.2.2.2 - ABR: WAN leg in Area 0, LAN VLANs in Area 1
- Summarized into Area 0 as: 172.20.100.0/22 (area 1 range 172.20.100.0 255.255.252.0)
- All PCs and Laptop-Guest-A on DHCP (Laptop-Guest-A received 172.20.103.2)

5. IP Addressing - Clinic B (R-ClinicB, Area 2)

VLAN	Department	Subinterface	Subnet	Gateway
10	Reception	Gi0/0/0.10	172.20.104.0/24	172.20.104.1
20	Doctors	Gi0/0/0.20	172.20.105.0/24	172.20.105.1
30	Nursing	Gi0/0/0.30	172.20.106.0/24	172.20.106.1

70	Guest Wi-Fi	Gi0/0/0.70	172.20.107.0/24	172.20.107.1
99	Native / switch mgmt	(trunk native)	-	-

- Serial0/1/0: 172.20.255.6/30 (DTE) toward R-MAIN
- OSPF Router ID: 3.3.3.3 - ABR: WAN leg in Area 0, LAN VLANs in Area 2
- Summarized into Area 0 as: 172.20.104.0/22 (area 2 range 172.20.104.0 255.255.252.0)
- All PCs and Laptop-Guest-B on DHCP (Laptop-Guest-B received 172.20.107.2)

6. Wireless Networks

Site	SSID	Security	Channel
Main Hospital	Willowbrook-Guest	WPA2-PSK / AES	6
Clinic A	Willowbrook-Guest-A	WPA2-PSK / AES	1
Clinic B	Willowbrook-Guest-B	WPA2-PSK / AES	11

7. Access Control - Permissions Granted and the Reasoning Behind Each

Every ACL below reflects the network's actual, final configuration (verified with show access-lists and live ping/SSH tests), not just the original design intent.

7.1 Remote Management (VTY) - Applied identically on R-MAIN, R-ClinicA, R-ClinicB

Control	Detail
Permission granted	access-list 10 permit 172.20.60.0 0.0.0.255 (Admin VLAN only), applied via access-class 10 in on all VTY lines (0–15); transport input ssh; login local
Reasoning	Only the hospital's IT administration team should be able to remotely manage any router, from any site. Since IT is centralized at the main hospital, one trusted subnet is permitted everywhere rather than maintaining a separate admin list per site.
Verified result	SSH from PC-Admin (172.20.60.x) succeeded on all three routers. SSH from PC-Reception (a non-admin subnet) timed out, confirming the restriction is enforced.

7.2 Guest Wi-Fi Isolation - GUEST-ISOLATION, applied per-site on each Guest subinterface, inbound

Control	Detail
Permission granted	deny ip <local Guest subnet> 172.16.0.0 0.15.255.255 (blocks the entire internal supernet); permit ip <local Guest subnet> any (guests may still reach the internet); deny ip any any
Reasoning	Patients and visitors have no legitimate reason to reach any clinical, administrative, or server subnet, at their own site or any other. The wildcard 172.16.0.0/12 was used because it fully covers the 172.20.0.0/16 address space actually in use (172.16.0.0/12 spans 172.16.0.0–172.31.255.255), so one rule protects every internal subnet without listing them individually.
Verified result	Guest devices at all three sites failed to reach the EHR server, staff PCs, and each other across sites (Destination host unreachable). This was the intended, correct result.

7.3 Electronic Health Records Access - EHR-ACCESS, applied on R-MAIN's Server subinterface (Gi0/0/0.200), outbound

Control	Detail
Permission granted	permit tcp/icmp from Doctors, Nursing, and Pharmacy at the main hospital, and Doctors and Nursing at both clinics, to the EHR server (172.20.200.10) on TCP/443 and ICMP; deny ip any host 172.20.200.10; permit ip any any (everything else unaffected)
Reasoning	Only staff with a direct clinical need, doctors, nurses, and pharmacy, should reach patient records. Reception and Administration were deliberately excluded since neither role requires direct EHR access. ICMP was explicitly permitted for authorized departments purely to make the restriction demonstrable and testable in the lab; a production deployment would likely omit ICMP, since ping is not operationally required for an EHR system.
Verified result	PC-Doctor and cross-site clinic doctors/nurses successfully pinged the EHR server; PC-Reception and Guest devices were denied (Destination host unreachable, match counter incrementing on the deny line).

7.4 DMZ - Patient Portal Protection (applied on R-MAIN)

ACL	Applied	Permission & Reasoning
ACL111-DMZ-OUT	Gi0/0/0.90, inbound	Permits all outbound traffic sourced from the Portal server (172.20.90.0/24) toward any destination, since a public-facing server must be able to respond to any internet client.

ACL112-DMZ-IN	Gi0/0/0.90, outbound	Permits only TCP/443 (HTTPS) to the Portal server from any source, plus ICMP from the Admin subnet only. Everything else denied. Reasoning: the public should only ever reach the portal over HTTPS; ICMP is restricted to Admin so only IT can ping it for troubleshooting, general staff and the public cannot.
ACL121-ANTISPOOF	Se0/2/0 (ISP-facing), inbound	Denies any inbound packet claiming a source address from the internal supernet (172.16.0.0/12), loopback (127.0.0.0/8), or multicast (224.0.0.0/4); permits everything else. Reasoning: no legitimate packet from the internet should ever claim to originate from inside the hospital's own network, seeing one is a clear spoofing signal, so it is dropped at the true network edge before it can reach anything internal.

Verified result: Server-Portal was reachable from the simulated internet (R-ISP) on HTTPS-equivalent access while ICMP from R-ISP was correctly blocked; internal staff (Admin) could ping the portal while other departments could not (by design); and R-ISP could not reach any internal hospital subnet.

8. Lessons Learned

These are genuine issues encountered and resolved during the build, kept here in the order they occurred, since the debugging process itself is as instructive as the final working config.

8.1 The passive-interface default trap

R-MAIN's OSPF process uses passive-interface default plus explicit no passive-interface lines to un-mute only the WAN-facing serial interfaces, this stops OSPF Hellos from ever reaching department VLANs. When Clinic B's serial link (Se0/1/1) was added, the corresponding no passive-interface line was forgotten. Every other part of the configuration was correct (IP addressing, OSPF network statement, area assignment), yet the neighbor never formed. The fix was a single missing line. Takeaway: whenever a new interface is added to a router already running passive-interface default, check explicitly whether it needs to be un-muted, it's an easy step to forget precisely because everything else can be perfect.

8.2 A single-character OSPF typo produced a misleading symptom

A network statement was typed as 172.20.255.255 instead of 172.20.255.5, a typo, not a design error. The result looked identical to a genuine adjacency failure (the neighbor simply never appeared), which could easily have led to hours spent re-checking area assignments or interface states instead of the one line that was actually wrong. Fixed with no network <wrong> followed by the correct network statement. Takeaway: when a neighbor doesn't form despite everything else appearing correct, checking the exact network statement against the exact interface IP (character by character) should happen early, not last.

8.3 Why the WAN link belongs in Area 0, not the branch's local area

Placing each clinic's serial link in Area 0 (rather than Area 1/2) is what makes the clinic router a genuine ABR, one leg in the backbone, one leg in its own area. This is what allows area range summarization to work at all. It was tempting, when first questioning the design, to assume the WAN link should belong to the branch's own area; working through why it doesn't reinforced a real understanding of what an ABR actually is, rather than just following a template.

8.4 ACLs cannot filter traffic on an interface with no IP processing

In a router-on-a-stick design, the physical parent interface (Gi0/0/0) carries no IP address of its own, all addressing lives on its subinterfaces. An ACL applied to the bare parent interface (as ACL101/ACL102 initially were) does not filter anything, because `show ip interface` confirms 'Internet protocol processing disabled' on that interface. The fix was to remove those two ACLs entirely rather than relocate them, since the anti-spoofing ACL at the true network edge and the DMZ-specific ACLs on the correct subinterface already achieved the intended protection without redundancy.

8.5 Named ACLs append to the bottom, order matters

Adding new permit lines to an already-existing named ACL (which already ended in a `deny ip any any`) appended the new lines after that final deny, making them unreachable. `show access-lists <name>` was used to confirm the actual line order before assuming a rule was broken. Given time constraints, the fastest reliable fix was deleting and rebuilding the ACL cleanly in the correct order, rather than inserting at specific line numbers.

8.6 A failed test is not automatically a broken configuration

When PC-Doctor could not ping the Patient Portal server, the immediate assumption was a routing or ACL bug. In fact, ACL112 was working exactly as designed, it deliberately restricts ICMP to the Admin subnet only, and PC-Doctor is not in that subnet. The real fix was retesting from PC-Admin, not changing the configuration. Takeaway: before treating a failed test as a bug, check the ACL's actual permit conditions against exactly which device is being used to test, the network was correct; the test was not.

9. Summary of What This Network Demonstrates

- Multi-area OSPF with route summarization at each ABR, reducing three clinic subnets each to a single advertised block
- VLAN segmentation across three sites with consistent department-to-VLAN mapping
- DHCP services scoped per VLAN, with static addressing reserved for servers referenced by ACLs
- WPA2/AES-secured guest wireless, isolated from all internal subnets by ACL, at every site
- Role-based access to sensitive data (EHR) enforced by ACL, verified with live pass/fail tests, not just written as a policy

- A DMZ hosting a public-facing service, protected by outbound/inbound ACL pairs and anti-spoofing filtering at the true internet edge
- Centralized, restricted remote management (VTY/SSH) across all routers